Linear equations

1 Solve the following equations:

a $x + 6 = 15$

b $x + 12 = 9$

c $21 = x + 13$

d $x - 4 = 10$

e $x - 8 = -20$

f $x - 21 = -25$

g $7 = x - 1$

h $6 = -16 + x$

i $5x = 45$

j $12 = 3x$

k $-5x = 30$

l $\frac{x}{8} = 6$

2 For each of the following one-step equations, describe the step required to solve it:

a $\frac{x}{4.1} = 7.6$

b $4.1x = 94.3$

c $x - 5 = 4$

d $9x = 7 + 8x$

3 When 10 is added to a number, the result is 12. What is the number?

4 When a number is multiplied by 10, the result is 60. What is the number?

5 Solve the following equations:

a $7x + 14 = 0$

b $8x - 9 = 39$

c $\frac{x}{2} + 8 = 10$

d $-6 + 2k = 10$

e $4(x - 6) = 12$

f $\frac{x}{7} + 3 = -1$

g $\frac{x}{2} - 3 = 2$

h $-n - \frac{7}{8} = 3$

i $\frac{x + 9}{7} = 4$

j $x + \frac{1}{5} = \frac{3}{5}$

k $\frac{x}{6} = \frac{5}{3}$

l $\frac{7}{9} = \frac{4}{x}$

6 Solve the following equations:

a $\frac{n}{2} + \frac{n}{3} = 15$

b $2(4x + 4) = 56$

c $4(x - 5) + 3 = -41$

d $-\frac{6x}{5} + 5 = 35$

e $\frac{11x + 22}{8} = 33$

f $\frac{4x - 12}{3} = -24$

g $6x - 20 = x$

h $9x - 45 = 4x$

i $3(x + 6) = x$

j $5x + 10 = x + 30$

k $9x + 4 = 10x + 2$

l $9p - 6 = 3p + 24$

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m $10r + 8 = 74 - r$

n $4(5x + 1) = -3(5x - 5)$

o $5(2x + 2) = -3(4x - 5) + 5x$

Applications

7 The life expectancy, y, of a person born in Japan is approximated by the equation $y = 0.27x + 72$, where x is the number of years since 1970.

a Make x the subject of this equation.

b Find the value of x for when the life expectancy will be 95. Round your answer three decimal places.

c In what year will the life expectancy in Japan reach 95 years?

8 The amount of oil, y, imported to the United States from Canada in millions of barrels per day can be approximated by the equation $y = 0.106x + 1.37$, where x is the number of years since 2000.

a Solve this equation for x.

b Hence determine the number of years after 2000 that the approximate number of oil barrels imported from Canada per day will be 3.5 million. Round your answer to three decimal places.

c Hence, in what year will the approximate number of barrels imported from Canada be 3.5 million per day?

Technology

9 Solve the following linear equations using a graphics calculator, or other technology:

a $6x + 10 = 4x + 18$

b $3(x + 8) = x$

c $\dfrac{5x - 15}{4} = -20$

d $\dfrac{2x}{5} - 1 = -5$

e $\dfrac{5x}{4} + 3 = -22$

f $2(x - 5) + 1 = -21$

g $3(4x + 1) = -3(3x - 2) + 4x$

h $\dfrac{z}{5} + \dfrac{z}{4} = 36$

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